03/24/2026: Classic Diffusion Model

Reference: *Deep unsupervised learning using nonequilibrium thermodynamics.*

CSCI 546: Diffusion Models

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Lab Overview

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Course Logistics

Announcements on syllabus

Reminder: The syllabus is periodically updated! Always check the syllabus in the repo.

- The reading for Thursday has been shortened to Secs 20.1 and 20.2.
- The coding day for simple diffusion models has been moved up to next Tuesday.

Announcements on lightning chats (LC)

- LC assignments have been made. (See the syllabus.)
  - If there's an &, you're working together.
  - If there's a comma, there's some flexibility. I'd prefer you to work together (e.g. to save class time, to improve the quality of the product), but if you feel strongly about working separately, I'll likely make an exception. Please talk to your partner after class and let me know by Thursday.

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- Slide logistics
  - Please send a draft of LC slides ahead of time, ideally the night before. I want to make sure I can open the slides. I may also provide some feedback.
  - Please send final version by email sometime before class starts.

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- Content reminder
  - Could be anything... summary of the paper, your take on it, or a deep dive on some aspect of it (including background topic).
  - Have fun with it! Goal is to provide an opportunity to learn and share your learning!

Midterm Grades

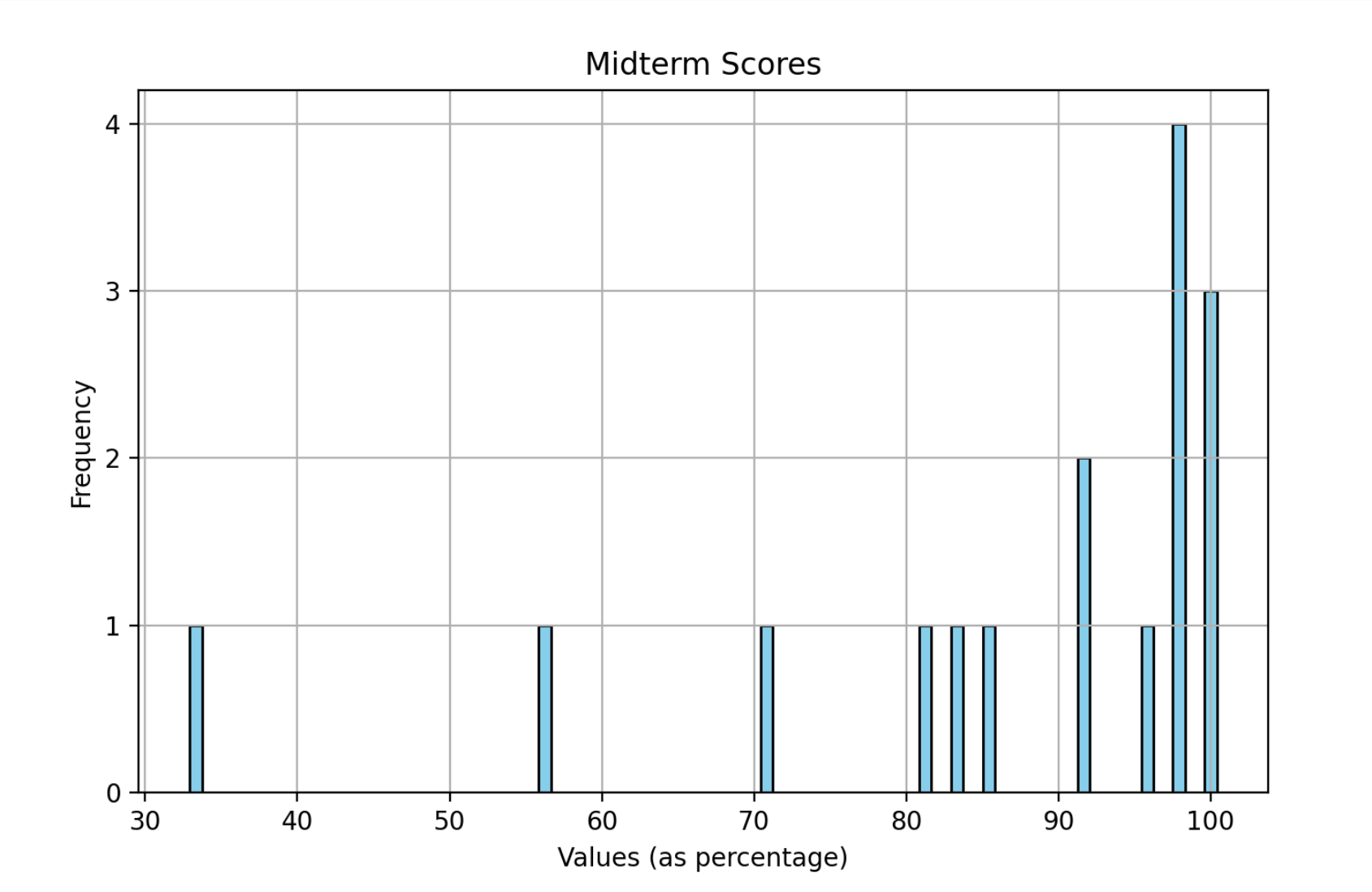

Figure 1: Median Score: 93.75%

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- If you are struggling with the material, please come to office hours or set up a 1-on-1 time. I want you to succeed! Truly!
- Canvas has been updated with all grades so far: midterm, group exercise participation, and 2 reading surveys.
- I will return this material now.

Reading Survey

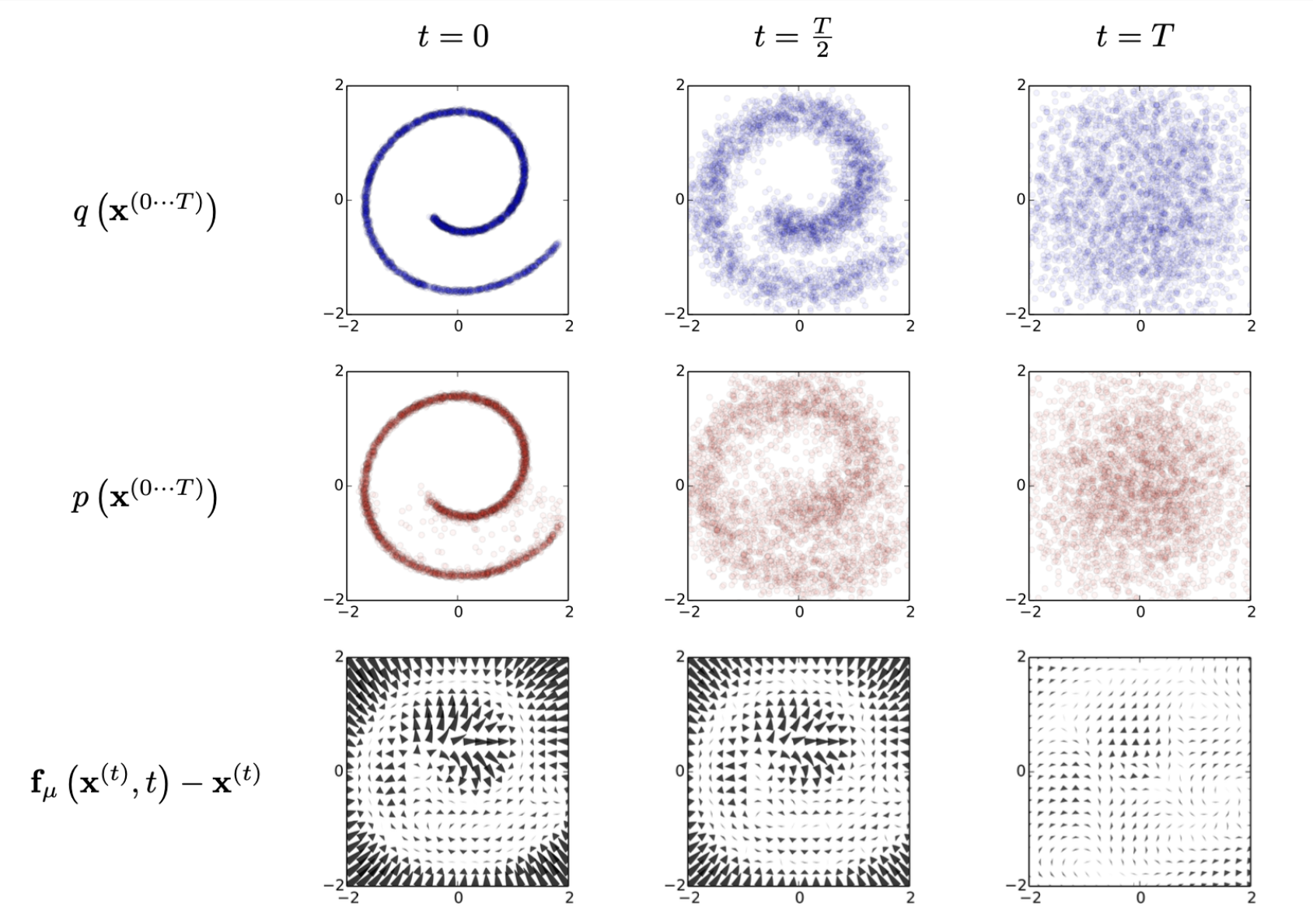
1. The paper proposes a method to flexibly learn complex probability distributions. Describe it.

Lightning Chat

Lightning Chat

Mini-Lecture

Gaussian Diffusion



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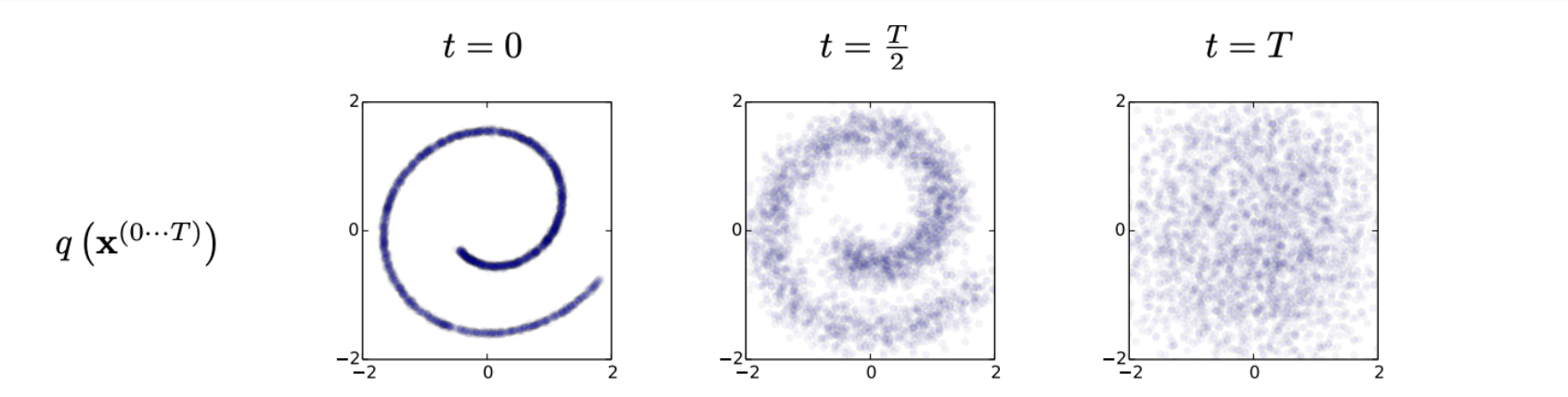

Forward trajectory.

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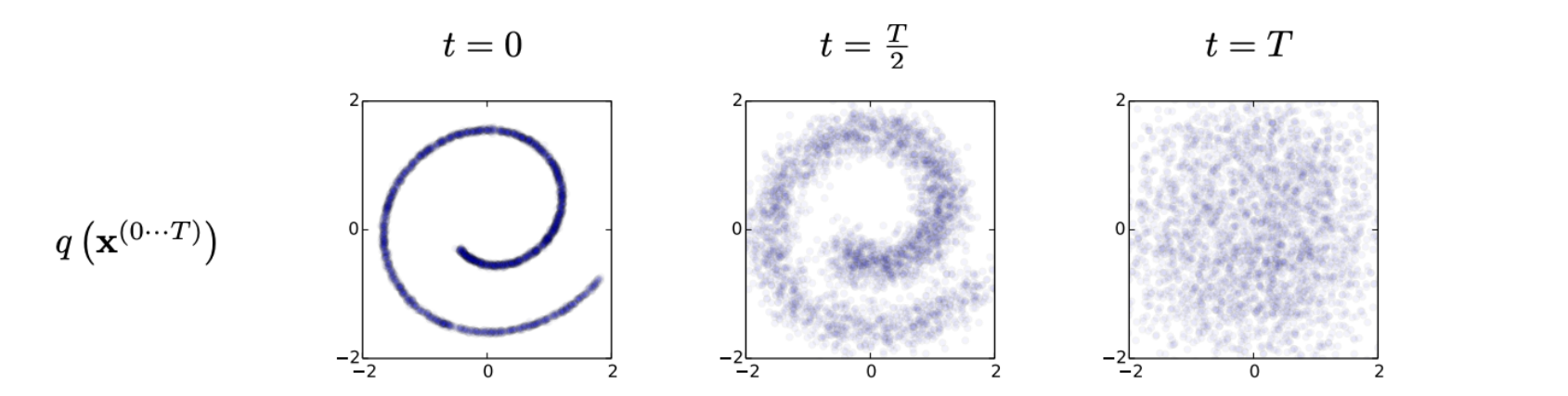

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Poll. What do these factors represent?

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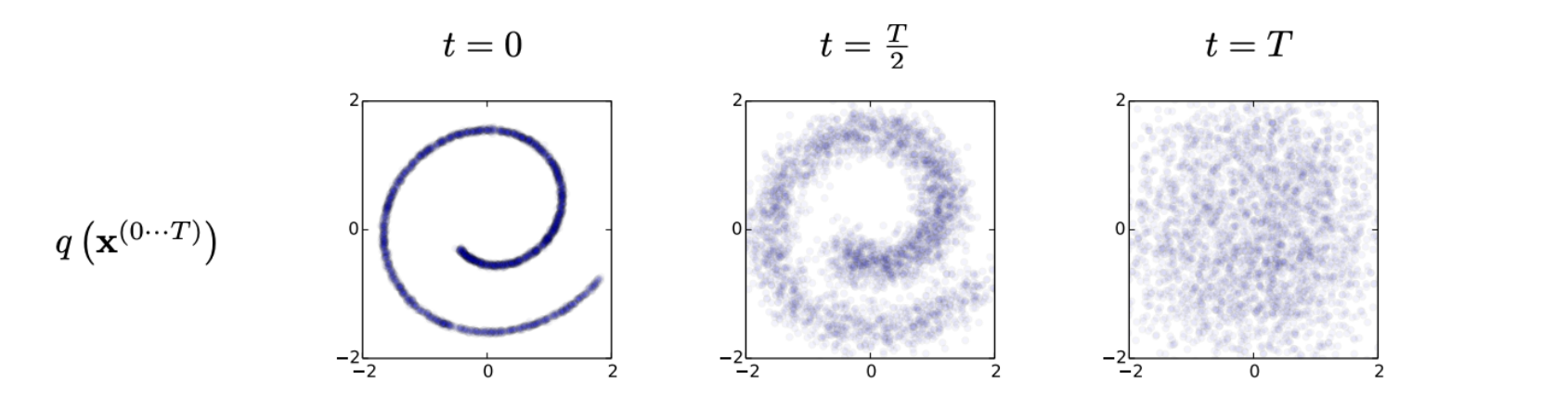

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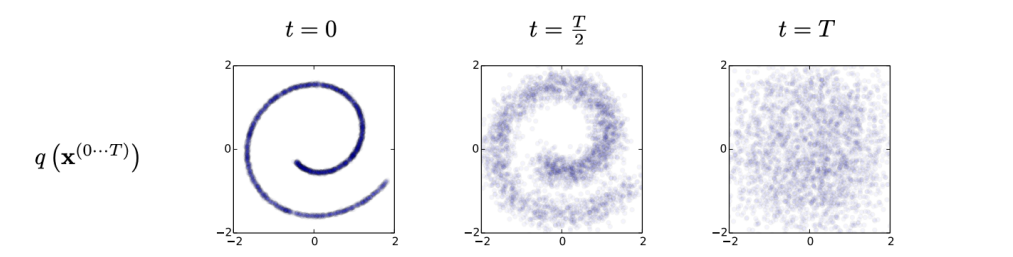

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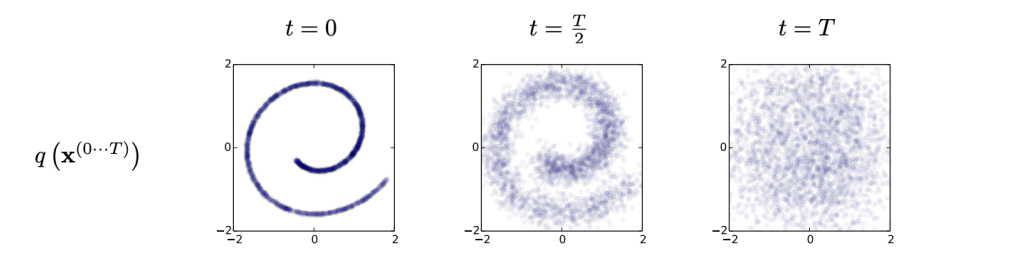

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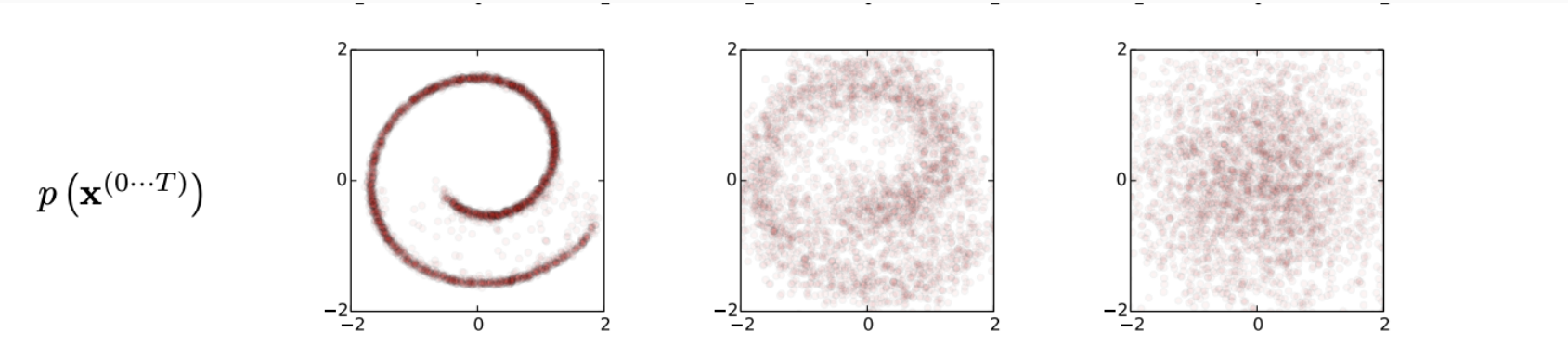

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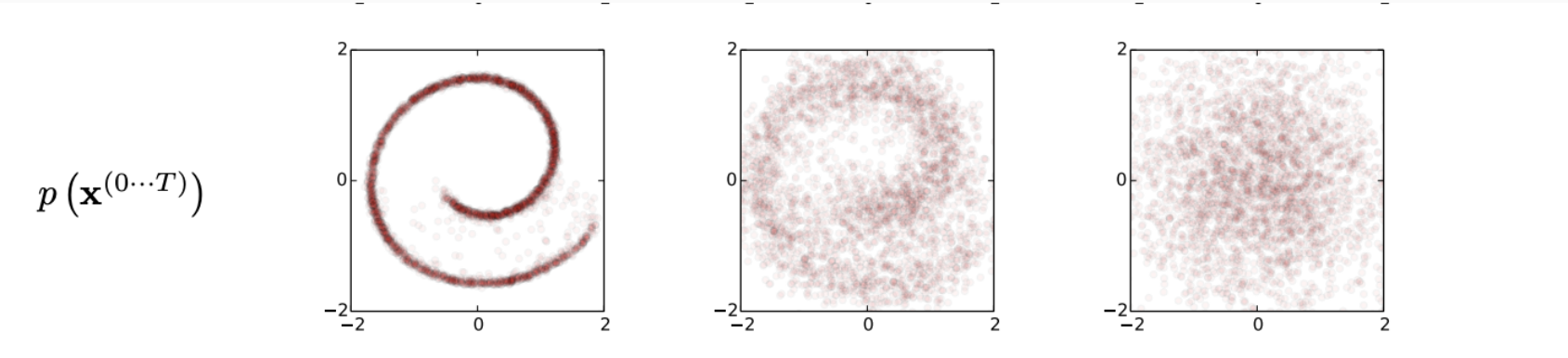

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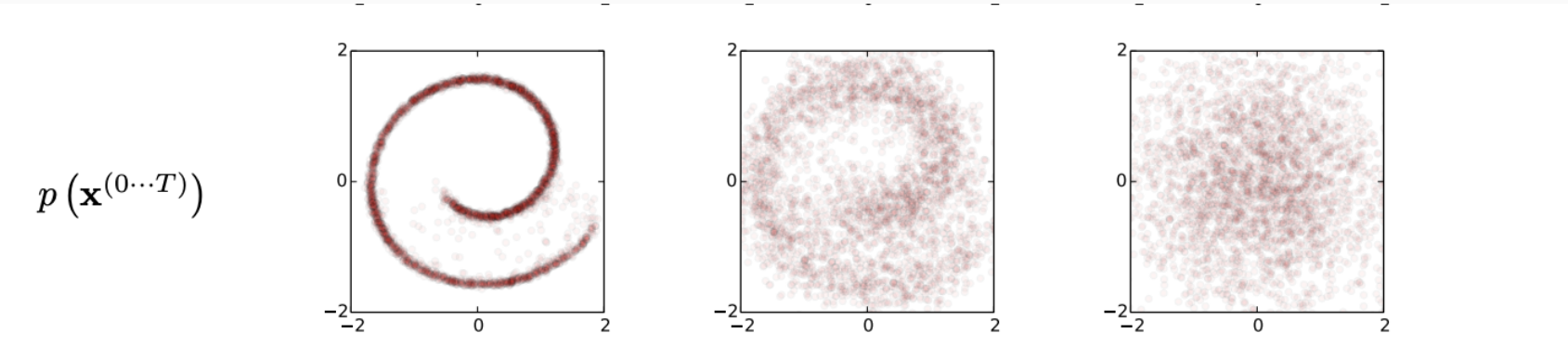

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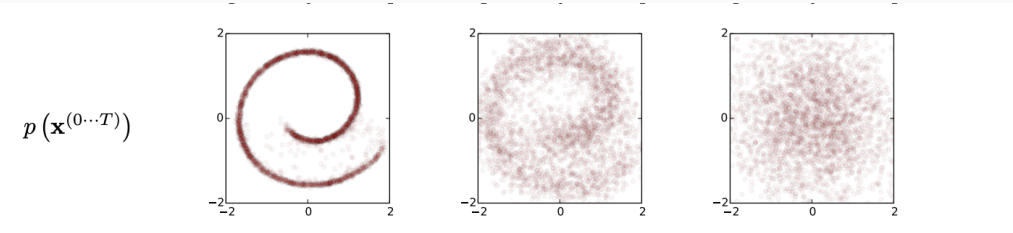

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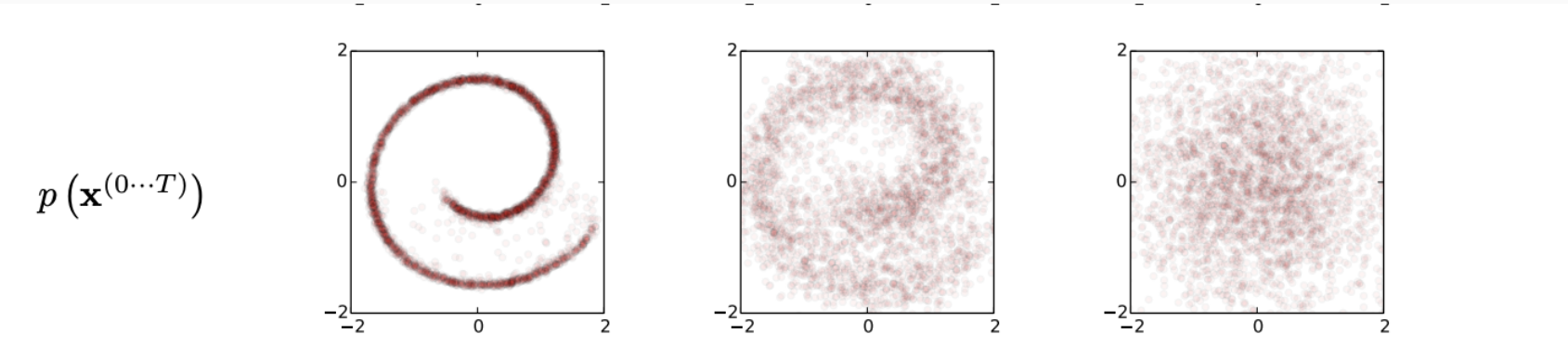

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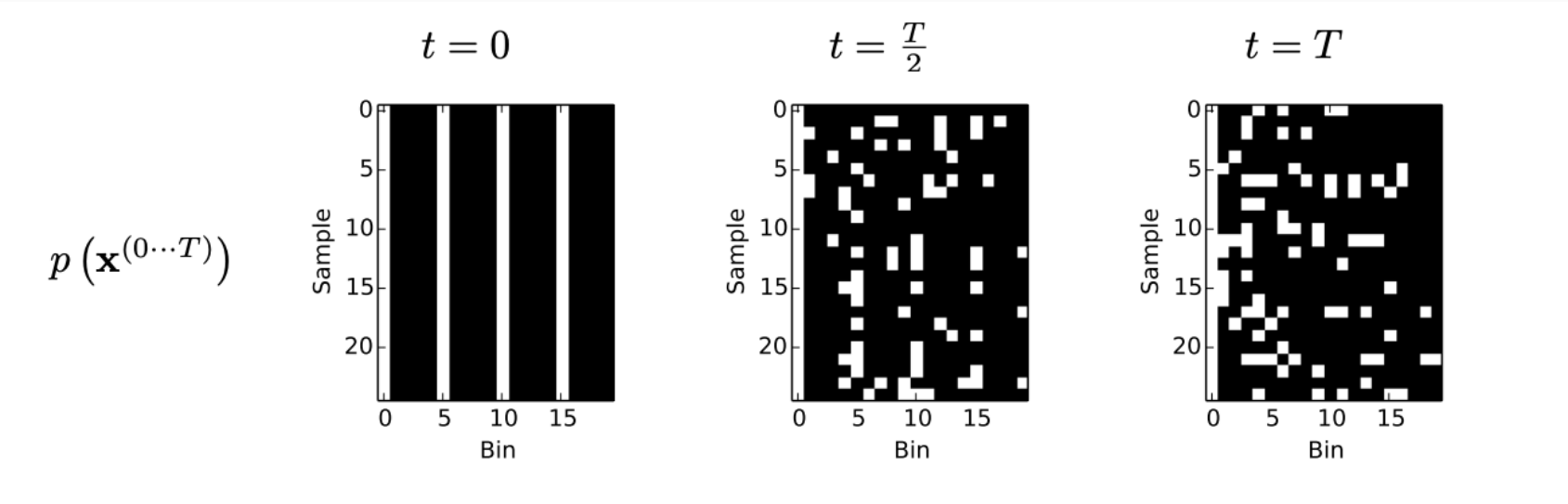

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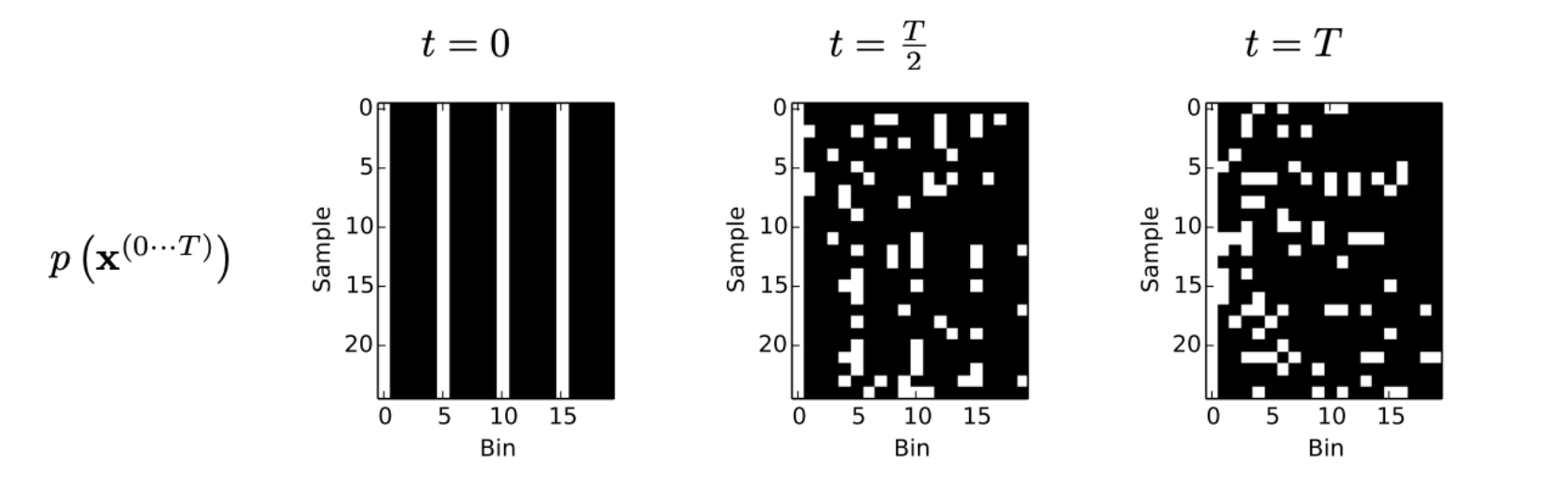

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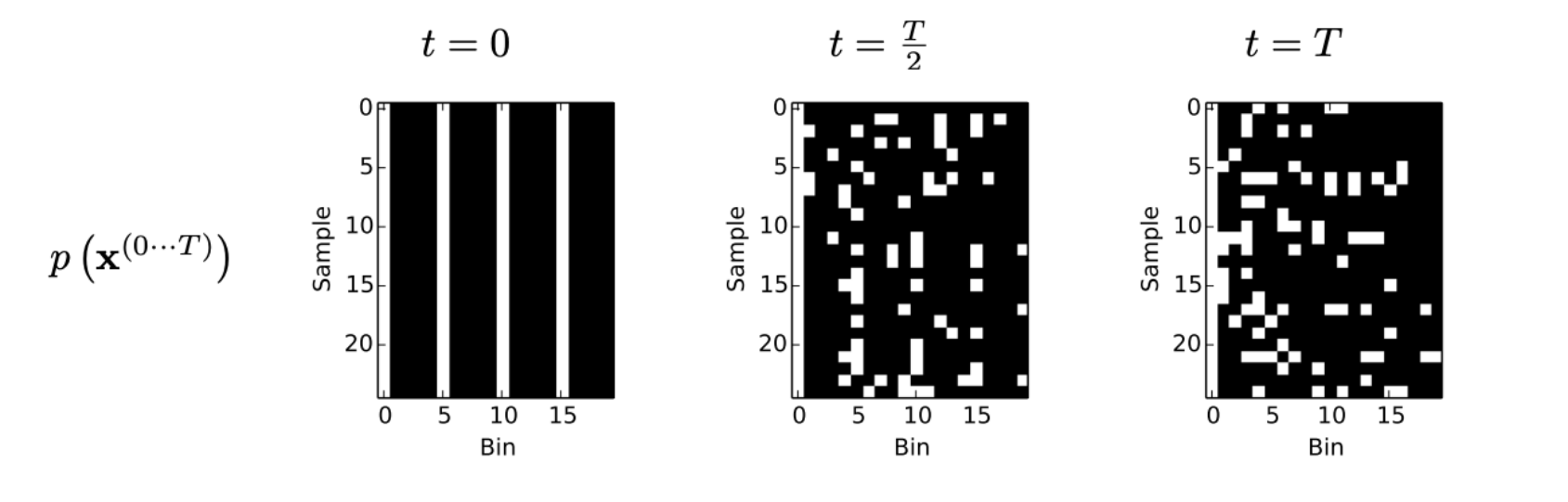

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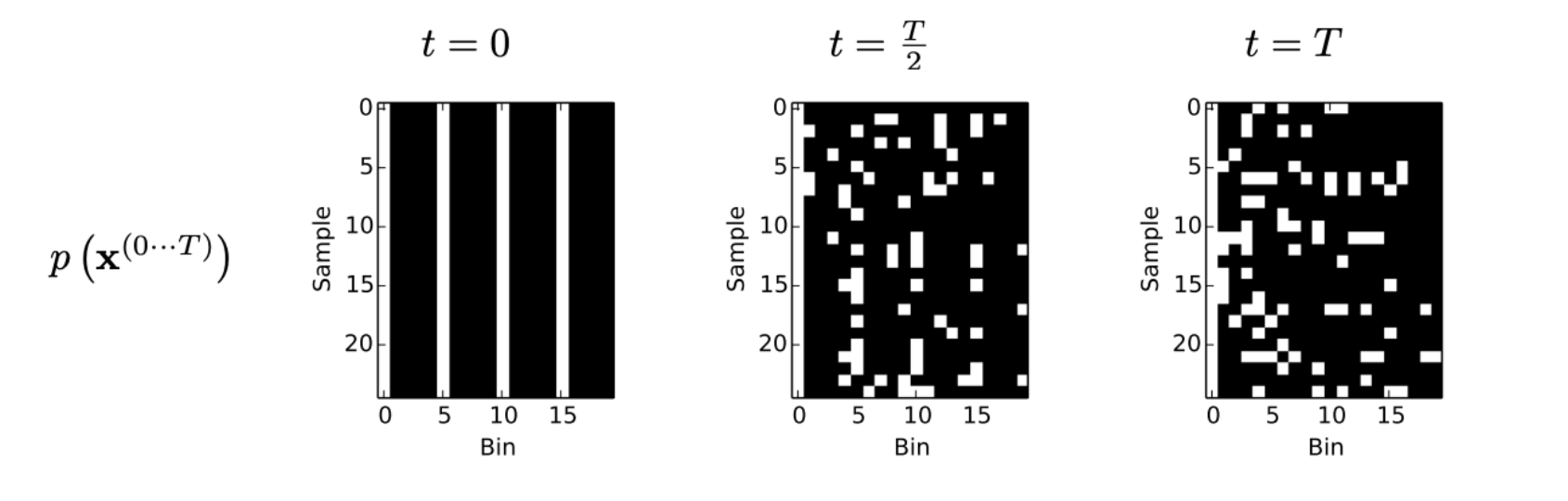

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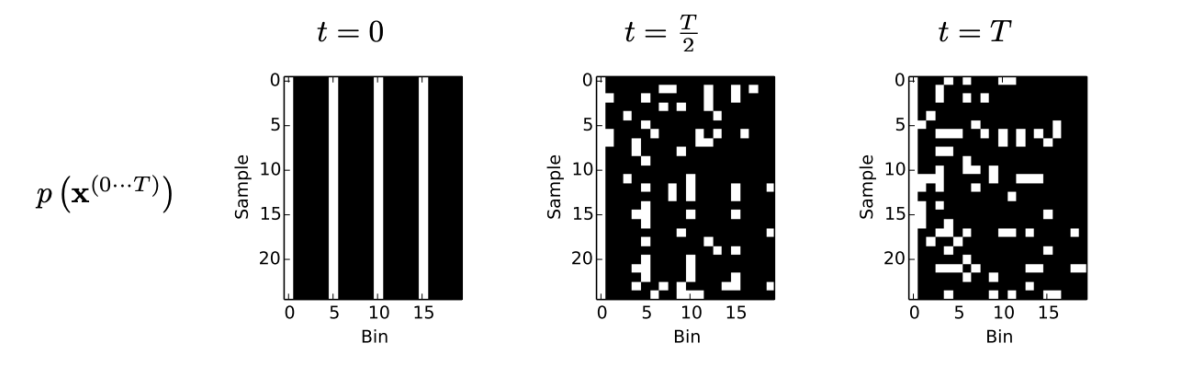

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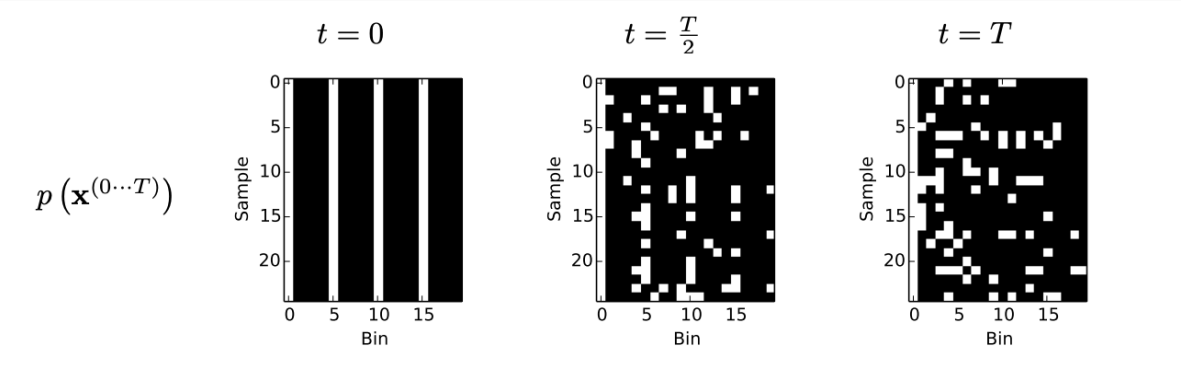

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So either way, we're "integrating" over possible starting values.

Group Work

Group Exercises

1. How do we know that if we define the forward process via the transition kernel $q(\boldsymbol{x}_t \mid \boldsymbol{x}_{t-1}) = \mathcal{N}(\boldsymbol{x}_t; \sqrt{1-\beta_t}\boldsymbol{x}_{t-1}, \beta_t \boldsymbol{I})$ that there is an equilibrium distribution, and moreover that the equilibrium distribution is $\pi = \mathcal{N}(0, \boldsymbol{I})$?

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Hint: (Assuming that the kernel is aperiodic and π-irreducible), it suffices to verify detailed balance:

That is, we must show

$$\pi(y)q(x \mid y) \stackrel{?}{=} \pi(x)q(y \mid x)$$

Good, Bad, Ugly, Beautiful

Guidelines Fill this out

1. *Good, Bad*: Technical content
2. *Ugly, Beautiful*: Presentational styles

Format

1. Small group discussion at whiteboards. (Fill out your responses there.)
2. Class discussion

Random groups

Aubrey Williams: 8
Austin Barton : 5
Blake Sigmundstad: 6
Diego Moylan: 7
Dillon Shaffer: 7
Ismoiljon Muzaffarov: 2
Jacob Tanner: 1
Josh Stoneback: 8
Joshua Bowen: 6
Joshua Culwell: 4
Laura Banaszewski: 4
Lina Hommel: 5
Logan Racz: 1
Matt Hall: 2
Mike Kadoshnikov: 2
Owen Cool: 3
Racquel Bowen: 1
Samuel Mocabee: 3
Tatiana Kirillova: 3